

AWS Cloud

AWS Cloud Fundamentals

- AWS Cloud Fundamentals

AWS Cloud Compute Services

- AWS Cloud Compute Services

Storage and Database Part-1

- Storage and Database Part-1

Storage and Database Part-2

- Storage and Database Part-2

Networking

- Networking

Security, Monitoring & Containers

- Security
- Monitoring
- Containers

Fundamentals - Programming Languages, GenAI Productivity Tools, etc.

SQL

- Introduction to SQL

Python for Data Engineers

- Introduction to Python
- Exploratory Data Analysis
- Data Processing and Visualization
- Python with SQL Database

Big Data

- Introduction to Big Data
- Data Lakes
- NoSQL

Introduction to AI, ML, DL, GenAI, RL & Agentic AI

- Introduction to AI, ML, DL, GenAI
- Introduction to Deep Learning
- RNNs and Transformers
- Convolutional Neural Networks
- Reinforcement Learning
- Agentic AI Fundamentals

Productivity Tools

- GitHub Copilot
- AI-Powered Productivity with Microsoft 365 Copilot

DW & Data Integration Fundamentals

Data Warehousing Fundamentals & Data Modeling Techniques

- Introduction to Data Warehouse
- Data Warehouse Characteristics
- OLTP vs OLAP
- DW Schemas and Objects
- ETL vs ELT
- Data Warehouse Architecture
- Introduction to Data Modeling

- Data Modeling Techniques
- Types of Data Models
- Types of Relationships
- Data Model Development Cycle
- Dimensional Data Modeling
- NoSQL Data Modeling

Introduction to Data Integration

- Introduction to Data Integration
- Introduction to Talend
- Working with Talend Components
- Creating and Deploying Jobs

PySpark

PySpark Refresher

- Introduction to Apache Spark
- Spark Cluster Architecture
- Spark Services
- Spark Ecosystem
- Distributed Data Processing using Spark
- Spark with Python
- Environment Preparation using Databricks
- RDD Concepts

Advanced PySpark

- PySpark DataFrames
- PySpark SQL
- PySpark Streaming

ETL and Synthetic Data using PySpark

- Introduction to Synthetic Data
- Synthetic vs Anonymized vs Real Data
- Creating Tabular Synthetic Datasets

- Creating Semi-Structured Synthetic Datasets
- Building a Sample Synthetic Dataset
- ETL Pipeline Development Using Synthetic Data
- End-to-End ETL Pipeline in PySpark
- Data Quality and Utility Validation
- Scaling Considerations and Best Practices

Databricks

Databricks Refresher

- Introduction to Databricks Platform
- Lakehouse Architecture
- Delta Lake
- Databricks Cluster Architecture
- Databricks File System (DBFS)
- Databricks Web Interfaces
- Data Ingestion into DBFS
- Databricks Assistant

Advanced Databricks - Building an Agentic Application

- Vector Search
- Retrieval Augmented Generation (RAG)
- Mosaic AI Agent Framework
- Implementing Agent Logic
- Unity Catalog Tool Functions
- Building and Deploying a Retrieval Agent

Advanced Databricks - Agentic Application Case Study

- Case Study: Build and Deploy Retrieval Agent

DevOps & Containerization Layer

DevDataOps Fundamentals

- DevOps
- DataOps

Version Control with GitHub

- GitHub Fundamentals

Version Control with Git

- Git Fundamentals

Release Management

- Build & Release Engineering Best Practices
- Release Management

Orchestration using GitHub Actions

- GitHub Actions

Container Orchestration using Docker and Kubernetes

- Docker Fundamentals
- Kubernetes Fundamentals

Microservices at Scale using AWS and Kubernetes

- Microservices Architecture
- AWS-based Microservices Deployment
- Kubernetes-based Microservices Deployment

Category	Module	Topics Covered
AWS Cloud	AWS Cloud Fundamentals	AWS Cloud Fundamentals

	AWS Cloud Compute Services	AWS Cloud Compute Services
	Storage and Database Part-1	Storage and Database Part-1
	Storage and Database Part-2	Storage and Database Part-2
	Networking	Networking
	Security, Monitoring & Containers	Security, Monitoring, Containers
Fundamentals - Programming Languages & GenAI Productivity Tools	SQL	Introduction to SQL
	Python for Data Engineers	Introduction to Python; Exploratory Data Analysis; Data Processing and Visualization; Python with SQL Database
	Big Data	Introduction to Big Data, Data Lakes, NoSQL
	Introduction to AI, ML, DL, GenAI, RL & Agentic AI	Introduction to AI, ML, DL, GenAI; Introduction to Deep Learning; RNNs and Transformers; Convolutional Neural Networks; Reinforcement Learning; Agentic AI Fundamentals
	Productivity Tools	GitHub Copilot; AI-Powered Productivity with Microsoft 365 Copilot
DW & Data Integration Fundamentals	Data Warehousing Fundamentals & Data Modeling Techniques	Introduction to Data Warehouse; DW Characteristics; OLTP vs OLAP; DW Schemas and Objects; ETL vs ELT; DW Architecture; Introduction to Data Modeling & Techniques; Types of Data Models; Types of Relationships; Data Model Development Cycle; Dimensional Data

		Modeling; NoSQL Data Modeling
	Introduction to Data Integration	Introduction to Data Integration; Introduction to Talend; Working with Talend Components; Creating and Deploying Jobs
PySpark	PySpark Refresher	Introduction to Apache Spark; Spark Cluster Architecture; Spark Services; Spark Ecosystem; Distributed Data Processing using Spark; Introduction to Spark using Python; Databricks Environment Setup; RDD Concepts
	Advanced PySpark	PySpark DataFrames & SQL; PySpark Streaming
	ETL and Synthetic Data using PySpark	Synthetic Data Fundamentals; Synthetic vs Anonymized vs Real Data; Creation of Tabular and Semi-Structured Synthetic Datasets; ETL Pipeline Development using Synthetic Data; Data Quality Validation; Scaling Best Practices
Databricks	Databricks Refresher	Databricks Platform; Lakehouse; Delta Lake; Cluster Architecture; DBFS; Databricks Web Interfaces; Data Ingestion into DBFS; Databricks Assistant
	Advanced Databricks - Building an Agentic Application	Vector Search; RAG; Mosaic AI Agent Framework; Implementing Agent Logic; Unity Catalog Tool Functions; Build and Deploy Retrieval Agent
	Advanced Databricks - Agentic Application Case Study	Case Study: Build and Deploy Retrieval Agent

DevOps & Containerization Layer	DevDataOps Fundamentals	DevOps; DataOps
	Version Control with GitHub	GitHub Fundamentals
	Version Control with Git	Git Fundamentals
	Release Management	Build & Release Engineering Best Practices; Release Management
	Orchestration using GitHub Actions	GitHub Actions
	Container Orchestration using Docker & Kubernetes	Docker and Kubernetes Fundamentals
	Microservices at Scale using AWS & Kubernetes	Microservices Architecture and Deployment on AWS & Kubernetes